A Vision-Based System Design and Implementation for Accident Detection and Analysis via Traffic Surveillance Video

Conception et mise en œuvre d'un système basé sur la vision pour la détection et l'analyse des accidents via les vidéos de surveillance du trafic

Yifan Sui, Shaodong Zhou, Zhiyang Ju, and Hui Zhang, *Senior Member, IEEE*

***Abstract*—In this work, we aim to investigate the problem of detecting and analyzing traffic accidents automatically and effectively through surveillance videos and implement the whole framework on an AI demo board. First, the technique of motion interaction field (MIF) that has the potential to detect crashes in a video is adopted to locate the crashed vehicles based on the interactions between multiple moving objects. Second, the YOLO v3 model is employed to identify the crashed vehicles within the appropriate location. In order to recover the vehicle trajectories before the collision, a hierarchical clustering approach is used, and the corresponding trajectories are obtained. Third, to facilitate the judgment of traffic police, the trajectory is projected to a vertical view by using a perspective transformation. The vehicle velocity is estimated accordingly with the unbiased finite impulse response (UFIR) approach that does not require statistical knowledge of the external noise. Then, the estimated velocity and the obtained collision angle from the vertical view can be utilized to analyze the traffic accident. Finally, to show the effectiveness and implementation performance of the proposed approach, an experiment is carried out based on a Huawei AI demo board named HiKey970 that is used for coding all the mentioned algorithms. Several accident surveillance videos act as the input of the demo board. The accidents are detected successfully, and the corresponding vehicle trajectories are recovered.**

***Résumé*—Dans ce travail, nous cherchons à étudier le problème de la détection et de l'analyse automatique et efficace des accidents de la route à partir de vidéos de surveillance et à mettre en œuvre l'ensemble sur un banc d'essai de l'IA. Tout d'abord, la technique du champ d'interaction de mouvement (MIF) qui a le potentiel de détecter les accidents dans une vidéo est adoptée pour localiser les véhicules accidentés en se basant sur les interactions entre plusieurs objets en mouvement. Ensuite, le modèle YOLO v3 est utilisé pour identifier les véhicules accidentés à l'endroit approprié. Afin de récupérer les trajectoires des véhicules avant la collision, une approche de regroupement hiérarchique est utilisée, et les trajectoires correspondantes sont obtenues. Troisièmement, pour faciliter le jugement de la police routière, la trajectoire est projetée sur une vue verticale en utilisant une transformation de perspective. La vitesse du véhicule est estimée en conséquence par l'approche de la réponse impulsionnelle finie sans biais (UFIR) qui ne nécessite pas la connaissance statistique du bruit externe. Ensuite, la vitesse estimée et l'angle de collision obtenu à partir de la vue verticale peuvent être utilisés pour analyser l'accident de la circulation. Enfin, pour montrer l'efficacité et les performances de mise en œuvre de l'approche proposée, une expérience est réalisée sur un banc d'essai de Huawei AI appelée HiKey970, utilisée pour coder tous les algorithmes mentionnés. Plusieurs vidéos de surveillance d'accidents servent d'entrée au banc d'essai. Les accidents sont détectés avec succès, et les trajectoires des véhicules correspondants sont récupérées.**

***Index Terms*—Accident detection, speed estimation, target tracking, unbiased finite impulse response (UFIR) filter, vehicles.**

I. Introduction

DURING the past decades, increasing attention has been paid to detect and analyze accidents via traffic surveillance equipment. The detection of crashes is mainly based on manual observation in the traffic management center (TMC). Although manual observation is reliable in most instances, it has a lot of shortcomings. On the one hand, it is difficult for people to detect all of the accidents in the entire city quickly, which means that the injured in a traffic accident may not be treated properly in many cases. On the other hand, the manual analysis of the cause of a traffic accident is sometimes inaccurate because it is hard to get the trajectory and speed through surveillance video. Thus, it is necessary to develop algorithms for automatically detecting and analyzing traffic accidents.

Several models based on deep learning [1], [2] have been presented for automatic traffic accident detection.

Manuscript received August 14, 2021; revised December 23, 2021; accepted February 14, 2022. Date of publication May 11, 2022; date of current version May 20, 2022. This work was supported in part by Defense Industrial Technology Development Program. *(Corresponding author: Hui Zhang.)*

Yifan Sui is with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China.

Shaodong Zhou, Zhiyang Ju, and Hui Zhang are with the School of Transportation Science and Engineering, Beihang University, Beijing 100091, China (e-mail: huizhang285@gmail.com).

Associate Editor managing this article's review: Khan A. Wahid.

Digital Object Identifier 10.1109/ICJECE.2022.3154294

These methods require training with large amounts of data and use complex neural networks to detect collisions in videos. However, limited by the amount of training data and high computational costs, these frameworks are difficult to be implemented practically. In addition, with the rise in the number of traffic surveillance videos, detecting and analyzing accidents throughout the whole city with a centralized system are difficult. It is necessary to build a distributed architecture consisting of embedded devices [3] deployed in every block of the city. Therefore, a lightweight framework that can be implemented on embedded devices is required.

In this article, we propose an accident detection and analysis framework that can be implemented on AI demo boards. Considering quick accident detection, a motion interaction field (MIF) model [4] is adopted to detect and localize traffic accidents. Regarding the analysis of traffic accidents, we use a YOLO v3 model and hierarchical clustering to get the trajectory of the vehicle before the collision. In order to analyze the accident accurately, we employ unbiased finite impulse response (UFIR) filtering and perspective transformation before estimating the speed and collision angle of vehicles in an accident. In addition, regarding the implementation of the designed system, we validate the framework on HiKey970 that is a Huawei AI demo board. This article is organized as follows. Section II discusses previous literature related to computer vision-accident detection, trajectory tracking, and the estimation of speed and collision angle. Section III introduces the fundamentals of the MIF model, YOLO v3, UFIR filter, and perspective transformation. Section IV shows the result of an experiment. Section V indicates the conclusions of the research.

II. Literature Review

A. Accident Detection

The methods of vision-based accident detection in the past two decades have been developed in three ways: modeling of traffic flow patterns, analyzing vehicle activities, and modeling of vehicle interactions [1]. In the first method, the typical traffic patterns are modeled based on traffic rules from large data samples. If the trajectory of a vehicle is inconsistent with typical trajectory patterns, it would be determined as an accident [5]–[7]. However, it is difficult to detect collisions because the trajectory data of collisions in the real world is limited. The second method detects accidents by calculating vehicle motion characteristics [8]–[10], such as speed, acceleration, and distance between two vehicles, which means that all vehicles need to be tracked continuously. As a result, the accuracy of the method in a crowded traffic environment is usually limited by computational capacity. In the third method, the interaction of vehicles is modeled with the application of the social force model [11] and the intelligence driver model [12]. This method requires a lot of training samples, and the performance is poor since it detects collision based on the change of vehicle speed only.

B. Vehicle Trajectory Tracking

Methods for vehicle tracking are mainly based on vehicle model, position, profile, and feature. The model-based algorithm makes 3-D models [13], [14] for real vehicles and matches them with vehicles in the videos. Although the result is accurate, this method induces a high computational load when there are many vehicles in the same video. The tracking speed of these methods based on vehicle position [15] or profile [16] is fast. However, the accuracy of the methods would decrease when vehicles are overlapped. For the tracking approach based on feature [17], [18], the performance is not satisfying if the video quality is not good.

Another issue for the vehicle tracking problem is the filtering of vehicle trajectories. The widely used approach for noise elimination is the Kalman filter (KF) [18]. Other improved algorithms of KF includes white noise model with adjustable level [19], variable dimension (VD) filtering [20], multiple model (MMs) [21], and interacting MMs (IMMs) [22]. However, the performance of KF and its improved algorithms depends too much on the accuracy of the statistical properties of noise and its prior information. However, the statistical properties of noise are hard to acquire in practice.

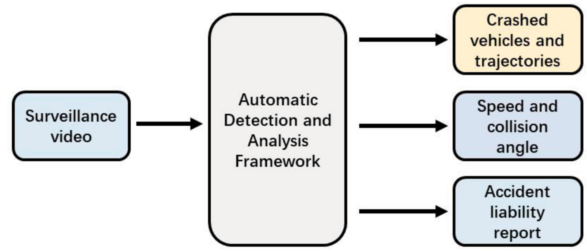

Fig. 1. Overall framework of the detection and analysis system.

C. Estimation of Speed and Collision Angle

In previous studies, the estimation of vehicle speed usually depends on radar [23], speed sensor [24], or other specialized equipment, such as digital aerial images [25] and stereo cameras [26]. Another vision-based method [27] uses a single camera above the road to get the vanishing point and the width of cars from the video, which means that it can only estimate the vehicle speed in a single direction. However, most traffic accidents happen at intersections, where vehicles drive in different directions.

Besides the speed, the estimation of collision angles is important for the accident analysis as well. Most of the existing methods [28]–[30] are based on specialized equipment when measuring the collision angle of cars. A vision-based method [28] for calculating angles in videos used a special camera and gets the distance between the object and camera in advance. A hybrid observer that is arranged on the four tires of a car to get the direction of vehicles was proposed in the literature [29]. These methods cost too much to put into practice.

III. Methods

In order to detect and analyze traffic accidents automatically from the surveillance videos, a comprehensive system is needed to identify the crashed vehicles, draw their trajectories, and estimate the speed and collision angle. The structure of the detection and analysis system is shown in Fig. 1.

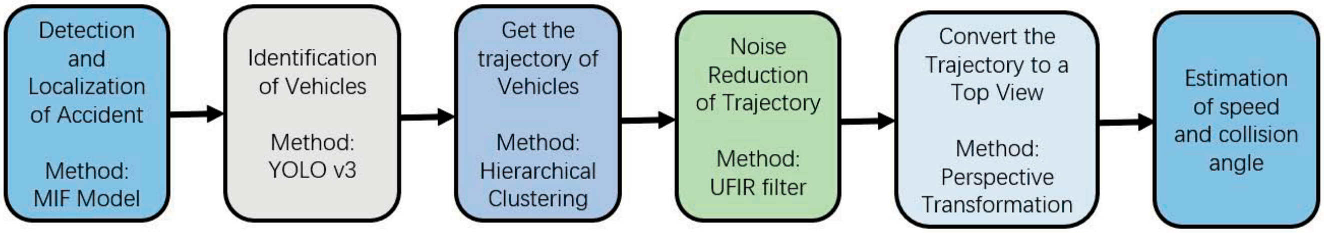

Fig. 2. Overall framework for detecting and analyzing traffic accidents.

Meanwhile, the framework is required to be implemented on the artificial intelligence demo board.

In this article, a framework for detecting and analyzing traffic accidents is created. As shown in Fig. 2, the overall framework consists of six major components. First, an MIF model is built to detect whether there are crashes in the video and identify when and where the crash occurs. After detecting and locating the accident, the vehicles in the accident are identified through YOLO v3. In order to analyze the accident, it is necessary to obtain the trajectory, speed, and collision angle of the accident vehicles. The trajectories of these vehicles are plotted using hierarchical clustering and UFIR filtering. Before measuring the speed and collision angle, the error caused by the angle of view of the monitoring probe shall be eliminated by converting the trajectories to a top view utilizing perspective transformation.

It is convenient for traffic police to detect and analyze an accident quickly with the help of the proposed framework in this article. Meanwhile, this framework is implemented completely on HiKey970, a Huawei HiSilicon artificial intelligence demo board. HiKey970 is one of the most advanced edge platforms in the world with a special neural network processing unit (NPU) whose performance in deep learning is 25 times higher than the CPU of the board. Using HiKey970, the proposed framework can make a contribution to the AI-based intelligent traffic camera in smart cities.

A. Motion Interaction Field

In order to detect an accident and locate the crashed vehicles, the MIF model is employed. MIF is a traffic detection model proposed by Yun [4]. The model is motivated by the movement of water waves when multiple objects are shifting on the water surface, pushing water around, and creating waves [4]. The MIF reflects the interaction between all moving objects in videos. The MIF is generated with Gaussian kernels after getting the speed and position of each moving point using an optical flow algorithm. After the filtering of MIF, a traffic accident can be detected and located if the maximum of MIF, which is called abnormality, exceeds the threshold. This method avoids complex training that needs huge training data and high computing power. Thus, it is suitable for our framework. Fig. 3 shows the overall scheme of the method.

The procedures of our method are summarized as follows.

Step 1 (Optical Flow): Get the speed $(\upsilon_{x_i}, \upsilon_{y_i})$ and position (x_i, y_i) of every object through the optical flow algorithm.

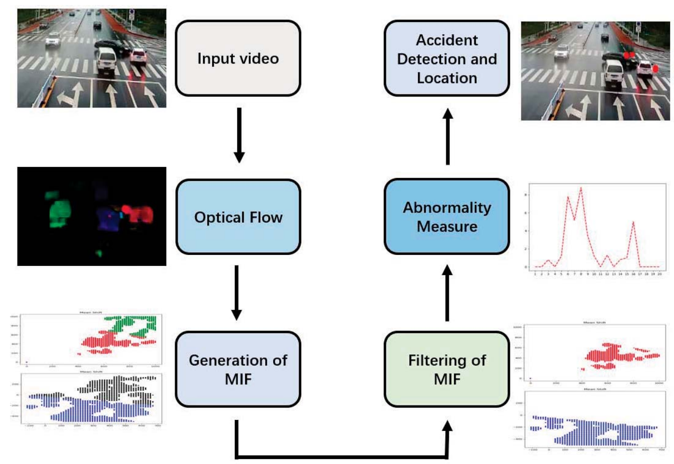

Fig. 3. Overall framework of the MIF model.

Step 2 (Generation of MIF): Calculate the kernel $k(x, y, x_i, y_i)$ by subtracting two Gaussian functions, which has different center positions. One center position is $(x_i + \upsilon_{x_i}, y_i + \upsilon_{y_i})$, showing the forward direction. The other is $(x_i - \upsilon_{x_i}, y_i - \upsilon_{y_i})$, which reflects the backward direction

$$K(x, y; x_i, y_i) = k\big(x, y; x_i + \upsilon_{x_i}, y_i + \upsilon_{y_i}\big) - k\big(x, y; x_i - \upsilon_{x_i}, y_i - \upsilon_{y_i}\big). \tag{1}$$

MIF is the sum of all kernels obtained in (1)

$$F(x, y) = \sum_{x_i, y_i} K(x, y; x_i, y_i) \tag{2}$$

where $F(x, y)$ refers to the MIF model.

*Step 3 (Filtering of MIF):*The MIF obtained through (2) is divided into positive regions and negative regions. After examining the one-to-one correspondence between each positive region and negative region through comparing the similarity of its Hu's image moment [31], the symmetrical regions will be removed. This is the filtering of the MIF model. Hu image moments are seven moments that show the similarity of two images. The formulas of Hu image moments are

$$\Phi_1 = \eta_{20} + \eta_{02}$$
$$\Phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$
$$\Phi_3 = (\eta_{20} - 3\eta_{12})^2 + 3(\eta_{21} - \eta_{03})^2$$
$$\Phi_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2$$

$$\begin{aligned}\Phi_5 &= (\eta_{30}+3\eta_{12})(\eta_{30}+\eta_{12})\left[(\eta_{30}+\eta_{12})^2-3(\eta_{21}+\eta_{03})^2\right]\\ &\quad+(3\eta_{21}-\eta_{03})(\eta_{21}+\eta_{03})\\ &\quad\times\left[3(\eta_{30}+\eta_{12})^2-(\eta_{21}+\eta_{03})^2\right]\\ \Phi_6 &= (\eta_{20}-\eta_{02})\left[(\eta_{30}+\eta_{12})^2-(\eta_{21}+\eta_{03})^2\right]+4\eta_{11}\\ &\quad\times(\eta_{30}+\eta_{12})(\eta_{21}+\eta_{03})\end{aligned}$$

$$\begin{aligned}\Phi_7 &= (3\eta_{21}-\eta_{03})(\eta_{30}+\eta_{12})\left[(\eta_{30}+\eta_{12})^2-3(\eta_{21}+\eta_{03})^2\right]\\ &\quad+(3\eta_{12}-\eta_{30})(\eta_{21}+\eta_{03})\\ &\quad\times\left[3(\eta_{30}+\eta_{12})^2-(\eta_{21}+\eta_{03})^2\right]\end{aligned}$$

where

$$\eta_{pq}=\frac{u_{pq}}{u_{00}^{\gamma}},\quad \gamma=(p+q+2)/2,\ \ p+q=2,3,\ldots$$
$$u_{pq}=\int_{-\infty}^{\infty}(x-\bar{x})^p(y-\bar{y})^q f(x,y)dxdy\quad p,q=0,1,2,\ldots$$

$f(x,y)$ is the image function and $(\bar{x},\bar{y})$ [31]. This is the filtering of the MIF model.

Step 4 (Abnormality Measure): Calculating the maximum of MIF through regions that have not been removed, we get the abnormal MIF intensity, which shows the degree of abnormality. The nonsymmetrical region with MIF intensity above the threshold is called the abnormal region, which shows the occurrence of the traffic accident in the video. The threshold is estimated through particle swarm optimization (PSO).

Step 5 (Detect and Locate the Accident): The coordinates of the abnormal region in MIF are proportional to that of the accident area in the video. Therefore, after getting the corresponding position of regions with intensity above the threshold, we can localize the accident through mapping the coordinates of these positions to the video frame.

B. YOLO v3

After the detection and location of crashed vehicles, the YOLO v3 model is utilized to identify the vehicles within the appropriate location. The YOLO network is a CNN network for the detection of objects. According to the unified size of traffic surveillance video, the size of our YOLO v3 network is 416×416, which balances recognition speed and accuracy. In order to avoid interference caused by other objects, the YOLO v3 model in this framework only identifies cars, motorbikes, and trucks. The picture to be detected is used as the input of the network, while the coordinates and probabilities of each bounding box can be generated directly through YOLO v3 [32]. The structure of the YOLO v3 network is illustrated as follows.

During training, the input image is divided into $S\times S$ grids. The grid whose result of prediction is true will predict a detected bounding box. The total number of detected bounding box is N_b. Each detected bounding box contains five parameters: x, y, w, h, and the confidence level. The (x,y) coordinate represents the center of the box. (w,h) means the weight and height of the box. The confidence level shows the confidence of the box in detecting objects. The calculation formula of the confidence level is

$$\text{confidence}=P_r(\text{Object})\times \text{IOU}_{b-\text{box}}^{\text{truth}}\tag{3}$$

where $P_r(\text{Object})$ means the possibility of containing object for the detected bounding box. $\text{IOU}_{b-\text{box}}^{\text{truth}}$ represent the accuracy of the detected bounding box.

Meanwhile, $P_r(\text{Class}|\text{Object})$ shows the probability that the object in the box belongs to a certain class. The class-specific confidence level calculated through (3) for each detected bounding box indicates the probability of classification. More details on the procedure can be seen in the YOLO v3 network shown in Fig. 4.

C. Hierarchical Clustering

Hierarchical clustering is a clustering algorithm that does not need to set the number of clusters in advance. It divides the datasets into different hierarchies. Each endpoint is in its own cluster at the beginning. The paired clusters are then merged into a single one, and the hierarchy is moved up until the termination condition is met [33]. Fig. 5 shows the process of hierarchical clustering. In the proposed framework, the image of the accident vehicle in each frame is used as the dataset of hierarchical clustering. We can get the trajectory from the clustering result.

The steps of hierarchical clustering are given as follows.

Step 1: Calculate the distance between each endpoint

$$\text{distance}_1=\sqrt{\sum_i(a_i-b_i)^2}\tag{4}$$

where a, b is the coordinate of an endpoint.

Step 2: Compare the distance between every two endpoints with (4). Merge the two nearest clusters into a single one.

Step 3: Calculate the distance between each cluster. The distance of two clusters is the distance between the two closest endpoints in the two clusters

$$\text{distance}_2=\sqrt{\sum_i(x_i-y_i)^2}\tag{5}$$

where x, y is the coordinate of two closest data points in each cluster.

Step 4: Compare the distance between every two clusters with (5). Repeat Step 2 and Step 3 until all endpoints are in one cluster.

D. UFIR Filtering

For all bounded inputs into the UFIR filter, the output is bounded. The UFIR filter can work without any information about noise and ensure that the phase–frequency characteristic of the output digital signal remains strictly linear after a digital signal with arbitrary amplitude–frequency characteristics is input. Meanwhile, the unit impulse response of the UFIR filter is finite. Therefore, the UFIR filter is suitable for our framework to make the trajectory smooth.

Suppose that the discrete kinematic model of the tracked vehicle is expressed as

$$\begin{aligned}\dot{x}_{n+1}&=F_n x_n\\ z_n&=H_n x_n\end{aligned}\tag{6}$$

where $x_n\in\mathbb{R}^K$ denotes the states, i.e., position, velocity, and acceleration, of the concerned vehicle at time step n, F_n is the

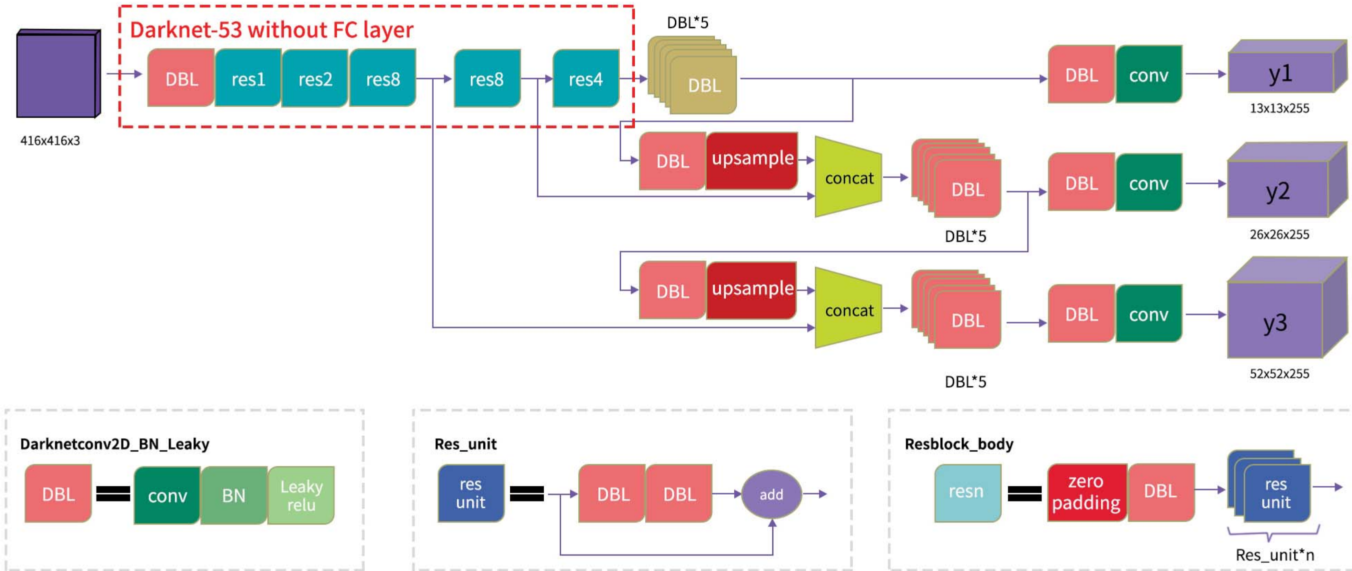

Fig. 4. YOLO v3 network.

state transmission matrix, $z_n \in \mathbb{R}^M$ is the measurement that is the vehicle's position in this article, and H_n is the measurement matrix. The UFIR filter is, thus, built based on this motion model of the vehicle.

The length of the UFIR filter window is assumed to be N, and the window is denoted as $[m, n]$ at time step n with $m = n - N + 1$. To ensure the causality of the UFIR filter, it requires that $m \geq 0$, i.e., $n \geq N - 1$. The estimation of the states of the target at time step n moment is represented as

$$\hat{\boldsymbol{x}}_n = \mathcal{F}_{n,0}^{m+1} \mathcal{H}_{n,m}^{-1} \boldsymbol{Z}_{n,m} \tag{7}$$

where

$$\begin{aligned}
\mathcal{F}_{r,h}^{r-g} &= \prod_{i=h}^{g} \boldsymbol{F}_{r-i} = \boldsymbol{F}_{r-h} \boldsymbol{F}_{r-h-1} \cdots \boldsymbol{F}_{r-g} \\
\mathcal{H}_{n,m}^{-1} &= \left(\boldsymbol{H}_{n,m}^{\mathrm{T}} \boldsymbol{H}_{n,m}\right)^{-1} \boldsymbol{H}_{n,m}^{\mathrm{T}} \\
\boldsymbol{Z}_{n,m} &= \left[\, z_n^{\mathrm{T}} \; z_{n-1}^{\mathrm{T}} \; \cdots \; z_m^{\mathrm{T}} \,\right]^{\mathrm{T}}
\end{aligned}$$

and

$$\begin{aligned}
\boldsymbol{H}_{n,m} &= \overline{\boldsymbol{H}}_{n,m} \boldsymbol{F}_{n,m} \\
\overline{\boldsymbol{H}}_{n,m} &= \mathrm{diag}(\boldsymbol{H}_n \quad \boldsymbol{H}_{n-1} \quad \cdots \quad \boldsymbol{H}_m) \\
\boldsymbol{F}_{n,m} &= \left[\mathcal{F}_{n,0}^{m+1\,T} \quad \mathcal{F}_{n,1}^{m+1\,T} \quad \cdots \quad \boldsymbol{F}_{m+1}^{T} \quad \boldsymbol{I}\right]^T
\end{aligned}$$

where $\boldsymbol{F}_n$ and $\boldsymbol{H}_n \in \mathbf{R}^{M \times K}$ are defined as in (6). Readers can refer to [34]–[36] for a detailed description of the UFIR filter.

E. Perspective Transformation

Perspective transformation is the conversion of a 3-D object in the space coordinate system into a 2-D image [37]. The principle of perspective transformation is shown in Fig. 6.

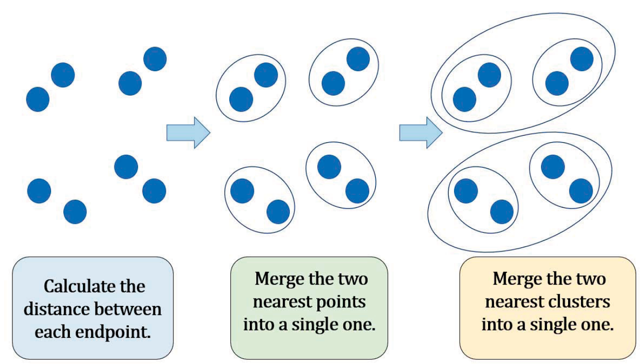

Fig. 5. Process of hierarchical clustering.

Define the P plane as the front view plane of an object and define point E as the view point. Perspective transformation is the process of transforming each pixel on the P plane to the corresponding pixel on the P' plane. The perspective transformation formula of the 2-D image is

$$\begin{aligned}
u &= \frac{ax + by + c}{gx + hy + 1} \\
v &= \frac{dx + ey + f}{gx + hy + 1}.
\end{aligned} \tag{8}$$

The (x, y) coordinates represent the position of a pixel on the P plane. The (u, v) coordinates represents the position of the corresponding pixel on the P' plane, and a, b, c, d, e, f, g, and h are the distortion parameters.

The distortion parameters a, b, c, d, e, f, g, and h are determined if the coordinates of four pixels on the P plane and the corresponding pixels on the P' plane are obtained using (8). Then, all pixels on P plane can be transformed to P' plane with (8).

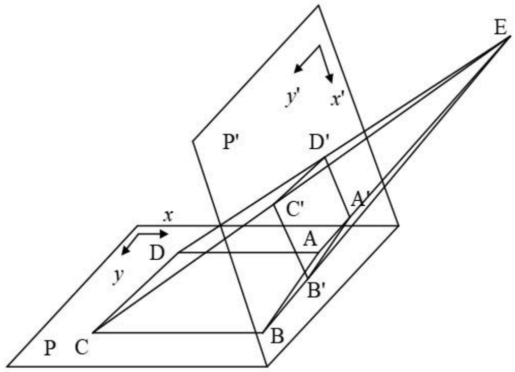

Fig. 6. Principle of perspective transformation.

F. Estimation of Speed and Collision Angle

Before estimating the vehicle speed, the distance between each pixel must be calculated. In our framework, we select the stop line in the surveillance video as the calibration to solve the perspective transformation equation. The specification of the stop line is 10.5 (m) $\times$ 0.15 (m). Therefore, the speed and collision angle of vehicles in an accident can be estimated after calculating the distance between each pixel in the vertical view. The process of estimation is shown in Fig. 7.

IV. Experimental Evaluation

In order to evaluate the performance, we have collected five videos with sudden traffic accidents [44], [45] as the dataset for the validation of the proposed approach. The results of the analysis are shown in Fig. 8. In the experiment, we chose one of the videos to show the effectiveness and implementation performance of the method. Meanwhile, a hardware practice test is conducted on HiKey970, a Huawei HiSilicon AI demo board.

This framework uses YOLO v3 pretrained model based on the COCO dataset and only recognizes cars, motorbikes, and trucks. After testing on traffic surveillance videos, the accuracies of the YOLO v3-416 pretrained model and the YOLO v3-608 pretrained model are similar, but the speed of the former is 1.75 times faster than that of the latter. Therefore, the YOLO v3-416 model is used in the experiment.

In the video, a black car and a white car collided at the intersection. In order to validate the proposed framework, we detect and analyze the accident that happened in the video. The resolution of the video is 720 $\times$ 576, while the frame rate is 30.

Using the MIF model in the framework, a frame that contains an accident would be detected if the abnormal MIF intensity (abnormality) is above the threshold. Meanwhile, we can localize the accident through mapping the coordinate of the maximum of abnormality to the video frame. The result is shown in Fig. 9. In Fig. 9, the black car and the white car collided at the upper right corner of the intersection. The red points show the coordinates of abnormalities above the threshold, which is the location of an accident.

TABLE I
Performance of Accident Detection

	ADR(%)	**FAR**(%)
The proposed method	87.5	16.7
The deep learning method	75.0	22.2

In order to verify the effect of the proposed framework, our accident detection model is compared against the unsupervised traffic accident detection method [40], a deep learning approach trained on the A3D dataset. For comparison, we use eight accident videos, five accident-free surveillance videos, and five videos that are easy to be misjudged as the dataset to test the performance of both methods, and the result is shown in Fig. 9. The accident detection rate (ADR) and the false alarm rate (FAR) are used as indicators to measure the detection effect. ADR and FAR are widely applied in the research of accident detection. Their formulas are shown by (9). The result of the comparison is shown in Table I. According to the result of the comparison, the accuracy of the proposed method is higher than the deep learning method. In addition, the FAR of our method is lower, which means that it has a better performance in situations that are prone to misjudgment. Meanwhile, our method does not require GPU and has a lower computing requirement

$$\begin{aligned}\text{Accident Detection Rate} &= \frac{\text{Accidents detected}}{\text{All Accidents}}\\ \text{False Alarm Rate} &= \frac{\text{Patterns false alarm happens}}{\text{All patterns}}.\end{aligned} \tag{9}$$

After getting the location of the accident, we can identify the accident vehicles with the proposed framework using YOLO v3. Fig. 10 provides the result of identification. The YOLO v3 model detects all the vehicles in the videos. If the detected bounding box of a car contains the coordinates of points with abnormalities above the threshold (the red points), the car is identified as an accident vehicle.

After identifying the accident vehicle, the YOLO v3 model will acquire the images of all vehicles frame by frame, which is shown in Fig. 11(a), and map the position of each vehicle in the video with its image. Through hierarchical clustering, the images of each car are integrated as a class. Since the image of the accident vehicle has been known, we can automatically filter out the class of the accident vehicle from all classes. Fig. 11(b) provides the result of clustering and filtering. After clustering, the images of accident cars are filtered out from all cars that appear in the video. The trajectories of the accident car are obtained as the position of the accident vehicle can be obtained from the image. The trajectories are shown in Fig. 12.

Fig. 13(c) shows the effect of UFIR filtering. The broken line represents the unfiltered vehicle trajectory, while the cyan line represents the trajectory after UFIR filtering. It can be seen that UFIR reduces noise and improves the authenticity of the trajectory effectively. The filtered trajectories of the two crashed vehicles are shown in Fig. 13(a). The blue points

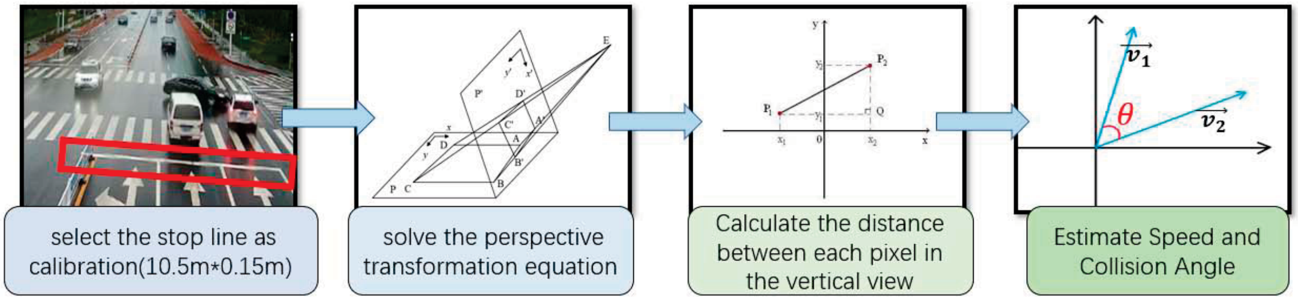

Fig. 7. Calibration of perspective transformation.

Fig. 8. Analysis result of the collected videos.

(a)

(b)

Fig. 9. Comparison of our method and deep learning method. (a) Detection and location of accident by our method. (b) Detection and location of accident by the deep learning method.

and red points are the coordinates of the two accident car trajectories. The two blue lines are the trajectories after UFIR filtering.

Then, using perspective transformation, the vertical view of filtered trajectory is shown in Fig. 13(b). The trajectories after perspective transformation accurately reflect the movement of the accident vehicles when the crash happened, which helps to analyze the cause of the accident. Then, the speed and collision angle of accident vehicles can be estimated. The effect of UFIR filtering and perspective transformation is shown in Table II. According to Table II, UFIR reduces the estimation error of speed by 19.12%. The estimation error of speed is reduced by 19.63%, while that of collision angle is reduced by 33.67% after using a perspective transformation.

Fig. 10. Identification of accident vehicles.

The accident liability report can be obtained through the framework. The speed of the black car was 35 km/h, while the maximum speed of the white car was nearly 90 km/h. On the one hand, the black car did not yield to the white car when turning, but, on the other hand, the white car exceeded the speed limit. Therefore, both sides should be responsible, and the main responsible part of the accident is the black car. According to the trajectories, the white car collided with the right-hand side of the black car at an angle of 63°. Compared with the traditional method that uses ground trajectory measurement, the proposed method is more convenient and accurate.

We conduct a hardware practice test on HiKey970, as shown in Fig. 14. HiKey970 is a Huawei AI demo board featuring

TABLE II
EFFECT OF UFIR AND PERSPECTIVE TRANSFORMATION

	Before UFIR	**After UFIR**	**After Perspective Transform**
The speed of white car	133.4km/h	107.9km/h	90.2km/h
The speed of black car	49.3km/h	37.6km/h	34.6km/h
The collision angle	47.2°	41.9°	63.2°

Fig. 11. Result of hierarchical clustering. The images of crashed vehicles, as shown in Fig. 11(b), are filtered out from the images of all cars in the video. (a) Before hierarchical clustering. (b) After hierarchical clustering.

Fig. 12. Trajectory of the accident vehicle.

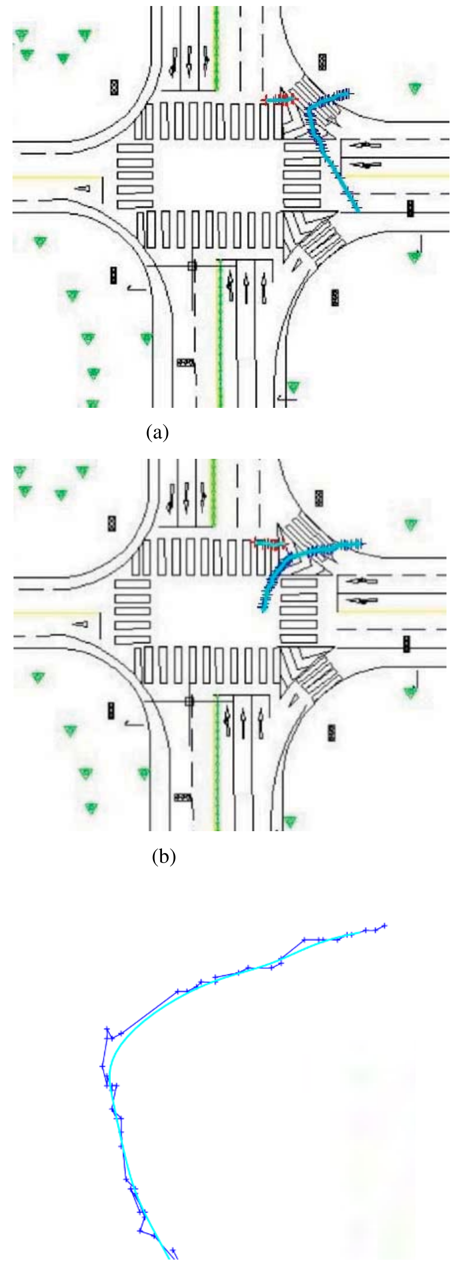

Fig. 13. Filtered trajectory of accident vehicles. (a) Before perspective transformation. (b) After perspective transformation. (c) Effect of UFIR filtering.

HiSilicon Kirin 970 SoC that adopts TSMC's 10-nm process. An ARM Cortex-A73 MPCore4 @ up to 2.36 GHz and an ARM Cortex-A53 MPCore4 @ up to 1.8 GHz are embedded as the CPU of the board [41]. Meanwhile, it contains an NPU for the acceleration of deep learning applications. Fig. 15 shows the structure of the board.

In order to verify the computing capability of Hikey970, the performance of the two platforms is compared in the experiment.

System 1: Intel Core i7-9750H CPU @ 2.60 GHz, 32-GB RAM.

TABLE III
DATASET COMPARISON

Platform	System1	System2
The speed of accident detection	9.8441s/Frame	21.5294s/Frame
The speed of vehicle identification	1.4857s/Frame	4.8371s/Frame
The speed of hierarchical clustering	37.1576s	143.7628s
The speed of UFIR filtering	6.1719s	23.1817s

System 2: HiKey970 with ARM Cortex-A73 MPCore4 @ up to 2.36 GHz and ARM Cortex-A53 MPCore4 @ up to 1.8 GHz, 6-GB RAM.

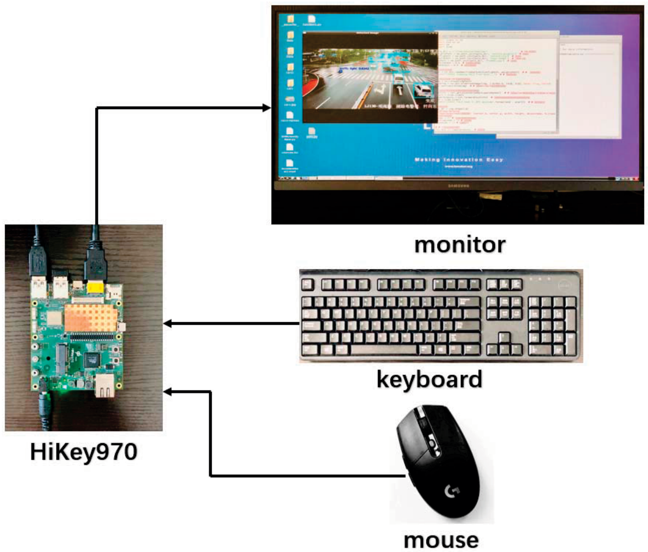

Fig. 14. Hardware practice test on HiKey970.

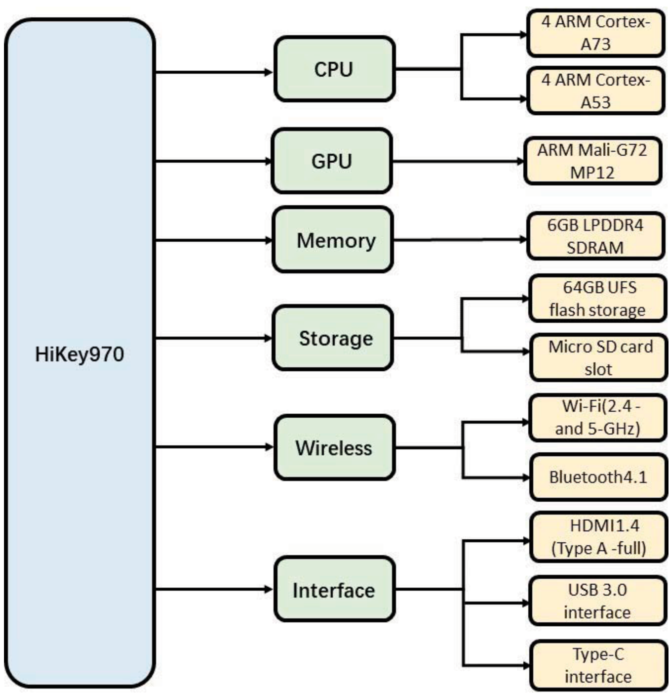

Fig. 15. Structure of HiKey970.

Table III shows the comparison of the two platforms. As shown in Table III, the calculation speed of HiKey970 is 45.72% of system 1 in accident detection, 30.71% in vehicle identification, 25.85% in hierarchical clustering, and 26.62% in UFIR filtering. The accuracy of the two platforms is identical because they execute the same algorithms. The speed of vehicle identification on HiKey970 using NPU will be around 6.5 times faster than that of now [42] because NPU has special optimization for deep learning.

Overall, the proposed framework can detect and analyze traffic accidents automatically through surveillance video. The vertical view of the trajectory can be seen clearly, while the speed and collision angle of the car can be estimated. Meanwhile, the framework performs well on both computer and HiKey970. However, there are still several limitations.

1) The identification accuracy of YOLO v3 will decrease if the vehicle is blocked by a large area.
2) The accident may be missed if the quality of the surveillance video is too low.
3) The NPU of HiKey970 is not used for acceleration in our experiment as Huawei has not released the API for executing a layer on NPU up until now [43].

V. Conclusion

In this article, a framework was proposed for detecting and analyzing traffic accidents automatically through surveillance video. First, the technique of the MIF model was utilized to detect and locate crashes in videos. Second, a YOLO v3 model was adopted for the identification of crashed vehicles. Third, the hierarchical clustering algorithm was used to recover the trajectories before the collision. In order to facilitate the judgment of traffic police, the trajectories were projected to a vertical view through perspective transformation. Using UFIR filtering, the trajectories were filtered, and the vehicle velocity was estimated. Then, an accident was analyzed by the estimated velocity and the obtained collision angle from the vertical view. Finally, a hardware practice test had been carried out for coding all the mentioned algorithms on HiKey970, a Huawei AI demo board. An accident surveillance video acted as the input of the demo board. The accident was detected successfully, and the corresponding vehicle trajectories were recovered. The performance of HiKey970 was 28.85%–45.72% of Intel Core i7-9750H CPU @ 2.60-GHz system.

However, there are still some problems to be solved in the further. First, another deep learning model can be tried to improve the identification accuracy when the car is blocked. Second, some image enhancement algorithms can be adopted for better performance of accident detection under different climate conditions or if the quality of surveillance videos is low. Third, the number plate of accident vehicles can be recognized for further analysis. In future research, we will focus more on path tracking control and attack detection for autonomous vehicles [44], [45].

References

[1] C. Regazzoni, A. Cavallaro, Y. Wu, J. Konrad, and A. Hampapur, “Video analytics for surveillance: Theory and practice,” *IEEE Signal Process. Mag.*, vol. 27, no. 5, pp. 16–17, Sep. 2010.

[2] X. Zhu, Z. Dai, F. Chen, X. Pan, and M. Xu, “Using the visual intervention influence of pavement marking for rutting mitigation—Part II: Visual intervention timing based on the finite element simulation,” *Int. J. Pavement Eng.*, vol. 20, no. 5, pp. 573–584, May 2019.

[3] C. F. Calvillo, A. Sánchez-Miralles, and J. Villar, “Synergies of electric urban transport systems and distributed energy resources in smart cities,” *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 8, pp. 2445–2453, Aug. 2018.

[4] K. Yun, H. Jeong, K. M. Yi, S. W. Kim, and J. Y. Choi, “Motion interaction field for accident detection in traffic surveillance video,” in *Proc. 22nd Int. Conf. Pattern Recognit.*, Aug. 2014, pp. 3062–3067.

[5] J. Varadarajan, R. Emonet, and J. Odobez, "Bridging the past, present and future: Modeling scene activities from event relationships and global rules," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2096–2103.

[6] T. Hospedales, S. Gong, and T. Xiang, "A Markov clustering topic model for mining behaviour in video," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Sep. 2009, pp. 1165–1172.

[7] W. Hu, X. Xiao, Z. Fu, D. Xie, T. Tan, and S. Maybank, "A system for learning statistical motion patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 9, pp. 1450–1464, Sep. 2006.

[8] S. Sadeky, A. Al-Hamadiy, B. Michaelisy, and U. Sayed, "Real-time automatic traffic accident recognition using HFG," in *Proc. 20th Int. Conf. Pattern Recognit.*, Aug. 2010, pp. 3348–3351.

[9] Y.-K. Ki, "Accident detection system using image processing and MDR," *Int. J. Comput. Sci. Netw. Secur.*, vol. 7, no. 3, pp. 35–39, 2007.

[10] D. Zeng, J. Xu, and G. Xu, "Data fusion for traffic incident detector using D-S evidence theory with probabilistic SVMs," *J. Comput.*, vol. 3, no. 10, pp. 36–43, Oct. 2008.

[11] R. Mehran, A. Oyama, and M. Shah, "Abnormal crowd behavior detection using social force model," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 935–942.

[12] W. Sultani and J. Y. Choi, "Abnormal traffic detection using intelligent driver model," in *Proc. 20th Int. Conf. Pattern Recognit.*, Aug. 2010, pp. 324–327.

[13] H.-N. Hu *et al.*, "Joint monocular 3D vehicle detection and tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5389–5398.

[14] A. Kuramoto, M. A. Aldibaja, R. Yanase, J. Kameyama, K. Yoneda, and N. Suganuma, "Mono-camera based 3D object tracking strategy for autonomous vehicles," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 459–464.

[15] H. Phat, D. Trong-Hop, and Y. Myungsik, "A probability-based algorithm using image sensors to track the LED in a vehicle visible light communication system," *Sensors*, vol. 17, no. 2, p. 347, 2017.

[16] J. Wu, "An automatic procedure for vehicle tracking with a roadside LiDAR sensor," *ITE J.*, vol. 88, no. 11 pp. 32–37, Nov. 2018.

[17] G. Lee, R. Mallipeddi, and M. Lee, *Trajectory-Based Vehicle Tracking at Low Frame Rates*. New York, NY, USA: Pergamon, 2017.

[18] K. Mantripragada, F. C. Trigo, F. P. Martins, and A. de Toledo Fleury, "Vehicle tracking using feature matching and Kalman filtering," in *Proc. 22nd Int. Congr. Mech. Eng. (COBEM)*. Hampton, IA, USA: ABCM, 2013, pp. 361–370.

[19] X.-J. Sun, Y. Gao, Z.-L. Deng, C. Li, and J.-W. Wang, "Multi-model information fusion Kalman filtering and white noise deconvolution," *Inf. Fusion*, vol. 11, no. 2, pp. 163–173, Apr. 2010.

[20] A. R. F. da Silva, "Bayesian mixture models of variable dimension for image segmentation," *Comput. Methods Programs Biomed.*, vol. 94, no. 1, pp. 1–14, Apr. 2009.

[21] B. Pourbabaee, N. Meskin, and K. Khorasani, "Sensor fault detection, isolation, and identification using multiple-model-based hybrid Kalman filter for gas turbine engines," *IEEE Trans. Control Syst. Technol.*, vol. 24, no. 4, pp. 1184–1200, Jul. 2016.

[22] W. Zhu, W. Wang, and G. Yuan, "An improved interacting multiple model filtering algorithm based on the cubature Kalman filter for maneuvering target tracking," *Sensors*, vol. 16, no. 6, p. 805, Jun. 2016.

[23] J.-S. Ho, P.-S. Liu, and K. Chen, "An unbiased average traffic speed estimation method for intelligent transportation system," in *Proc. 12th Int. Conf. ITS Telecommun.*, Nov. 2012, pp. 589–593.

[24] E. Odat, J. S. Shamma, and C. Claudel, "Vehicle classification and speed estimation using combined passive infrared/ultrasonic sensors," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 5, pp. 1593–1606, May 2018.

[25] H. Long, Y.-N. Chung, and J. De Li, "Automatic vehicle speed estimation method for unmanned aerial vehicle images," *J. Inf. Hiding Multimedia Signal Process.*, vol. 9, no. 2, pp. 442–451, 2018.

[26] P. Chanawangsa, J. Wan, C. Wu, and C. W. Chen, "A novel 2D–3D hybrid approach to vehicle trajectory and speed estimation," in *Proc. 17th Int. IEEE Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2014, pp. 1906–1907.

[27] C. Wang and A. Musaev, "Preliminary research on vehicle speed detection using traffic cameras," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2019, pp. 3820–3823.

[28] A. V. Bharadwaj, S. Paul, L. R. Kumar, and A. Somanathan, "An improved real-time approach for video based angular motion detection and measurement," in *Proc. IEEE Int. Conf. Comput. Intell. Comput. Res. (ICCIC)*, Dec. 2018, pp. 1–6.

[29] M. Shang, L. Chu, J. Guo, and Y. Fang, "Estimation of vehicle side slip angle using hybrid observer," in *Proc. 29th Chin. Control Conf.*, 2010, pp. 5378–5383.

[30] B. Zhang, H. Du, J. Lam, N. Zhang, and W. Li, "A novel observer design for simultaneous estimation of vehicle steering angle and sideslip angle," *IEEE Trans. Ind. Electron.*, vol. 63, no. 7, pp. 4357–4366, Jul. 2016.

[31] M.-K. Hu, "Visual pattern recognition by moment invariants," *IRE Trans. Inf. Theory*, vol. 8, no. 2, pp. 179–187, Feb. 1962.

[32] Y. Tian, G. Yang, Z. Wang, H. Wang, E. Li, and Z. Liang, "Apple detection during different growth stages in orchards using the improved YOLO-V3 model," *Comput. Electron. Agricult.*, vol. 157, pp. 417–426, Feb. 2019.

[33] L. Rokach and O. Maimon, *Clustering Methods*. Boston, MA, USA: Springer, 2005, pp. 321–352.

[34] Y. S. Shmaliy, "Linear optimal FIR estimation of discrete time-invariant state-space models," *IEEE Trans. Signal Process.*, vol. 58, no. 6, pp. 3086–3096, Jun. 2010.

[35] Y. S. Shmaliy, "An iterative Kalman-like algorithm ignoring noise and initial conditions," *IEEE Trans. Signal Process.*, vol. 59, no. 6, pp. 2465–2473, Jun. 2011.

[36] W. H. Kwon, P. S. Kim, and S. H. Han, "A receding horizon unbiased FIR filter for discrete-time state space models," *Automatica*, vol. 38, no. 3, pp. 545–551, Mar. 2002.

[37] A. Bonin-Font, A. Burguera, A. Ortiz, and G. Oliver, "Concurrent visual navigation and localisation using inverse perspective transformation," *Electron. Lett.*, vol. 48, no. 5, pp. 264–266, 2012.

[38] (Jun. 29, 2019). *Summary of Major Traffic Accidents in the Second Half of March 2019*. [Online]. Available: https://www.bilibili.com/video/av57174098/

[39] (Feb. 22, 2017). *Dafang Collection of Traffic Accidents in China-1612*. [Online]. Available: https://www.bilibili.com/video/av8762469/

[40] Y. Yao, M. Xu, Y. Wang, D. J. Crandall, and E. M. Atkins, "Unsupervised traffic accident detection in first-person videos," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Nov. 2019, pp. 273–280.

[41] D. Kang, J. Oh, J. Choi, Y. Yi, and S. Ha, "Scheduling of deep learning applications onto heterogeneous processors in an embedded device," *IEEE Access*, vol. 8, pp. 43980–43991, 2020.

[42] 96Boards. *Hikey970*. Accessed: Mar. 20, 2021. [Online]. Available: https://www.96boards.org/product/hikey970/

[43] F. Andrei. (Jan. 22, 2018). *Hisilicon Kirin 970—Android SoC Power & Performance Overview*. [Online]. Available: https://www.anandtech.com/show/12195/hisilicon-kirin-970-power-performance-overview/

[44] X. Xu, X. Li, P. Dong, Y. Liu, and H. Zhang, "Robust reset speed synchronization control for an integrated motor-transmission powertrain system of a connected vehicle under a replay attack," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5524–5536, Jun. 2021.

[45] J. Chen, Z. Shuai, H. Zhang, and W. Zhao, "Path following control of autonomous four-wheel-independent-drive electric vehicles via second-order sliding mode and nonlinear disturbance observer techniques," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2460–2469, Mar. 2021.

Yifan Sui received the B.Eng. degree in communication engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2021. He is currently pursuing the Ph.D. degree with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China.

His research interests include accident detection and signal processing.

Shaodong Zhou received the B.Sc. degree in transportation science and engineering from the Shandong University of Technology, Zibo, China, in 2015, and the M.Sc. degree in transportation engineering from Chang'an University, Xi'an, China, in 2018. He is currently pursuing the Ph.D. degree with the School of Transportation Science and Engineering, Beihang University, Beijing, China.

His research interests include intelligent systems, human–machine shared control for intelligent vehicles, and applications of machine learning algorithms.

Zhiyang Ju received the B.Eng. degree from the Harbin Institute of Technology, Harbin, China, in 2014, and the Ph.D. degree in electrical and electronic engineering from the University of Melbourne, Parkville, VIC, Australia, in 2019.

He currently holds a post-doctoral position at Beihang University, Beijing, China. He is also a Visiting Scholar with the Department of Mechanical Engineering, University of Melbourne. His research interests include cyber security of cyber–physical systems, multiagent systems, vehicle dynamics and control, and autonomous vehicles.

Hui Zhang (Senior Member, IEEE) received the B.Sc. degree in mechanical design manufacturing and automation from the Harbin Institute of Technology at Weihai, Weihai, China, in 2006, the M.Sc. degree in automotive engineering from Jilin University, Changchun, China, in 2008, and the Ph.D. degree in mechanical engineering from the University of Victoria, Victoria, BC, Canada, in 2012.

Dr. Zhang is also a member of the SAE International and American Society of Mechanical Engineers (ASME). He also serves as a Board Member of *International Journal of Electric and Hybrid Vehicles, Mechanical Systems*, and *Signal Processing*. He was a recipient of the 2017 IEEE TRANSACTIONS ON FUZZY SYSTEMS Outstanding Paper Award, the 2018 SAE Ralph R. Teetor Educational Award, the IEEE Vehicular Technology Society 2019 Best Vehicular Electronics Paper Award, and the 2019 SAE International Intelligent and Connected Vehicles Symposium Best Paper Award. He also serves as an Associate Editor for IEEE TRANSACTIONS ON INTELLIGENT VEHICLES, *The Journal of the Franklin Institute, SAE International Journal of Connected and Automated Vehicles*, IEEE OPEN JOURNAL OF THE INDUSTRIAL ELECTRONICS SOCIETY, and *ASME Transactions Journal of Dynamic Systems, Measurement and Control*.